UNIVERSITY ADMISSION DECISION ENGINE

Python Project Report

1.Overview

The University Admission Decision Engine is a system designed to evaluate student applications and determine admission eligibility based on predefined university rules. It considers factors such as entrance examination marks, academic percentage, and eligibility requirements.

The system automatically checks whether an applicant satisfies the admission criteria and generates a result such as Eligible, Not Eligible, or Further Review Required. It reduces manual work, minimizes processing errors, and improves the efficiency and consistency of the admission process.

2.Example Policy Used

The following sample admission policies are used for demonstration purposes.

1. The applicant must have completed the required qualifying examination.
2. The minimum academic percentage required is 50%.
3. The minimum entrance examination score required is 60 out of 100.
4. Applicants who satisfy both criteria are considered eligible for admission.
5. Applicants who fail to satisfy either minimum requirement are marked as not eligible under this sample policy.
6. Invalid or missing application details must be flagged for correction.
7. The system must display the final admission decision.
8. The final admission decision is subject to document verification and the university's official admission rules.

Example:

- Academic Percentage: 75%
- Entrance Exam Score: 80/100
- Qualifying Examination: Completed
- Decision: Eligible

Note: These are illustrative criteria only. Actual admission requirements vary by university and course.

3.Problem Statement

1. Manual admission processing requires considerable time and effort.
2. Evaluating a large number of student applications manually may lead to errors.
3. Applying admission eligibility criteria consistently can be challenging.
4. Incorrect calculations may result in inaccurate admission decisions.
5. Identifying eligible and ineligible applicants manually is time-consuming.
6. Universities need an efficient system to process applications and generate admission decisions accurately.

The proposed system addresses these problems by automating eligibility checks using predefined admission policies.

4.Objectives

1. To automate the university admission eligibility process.
2. To evaluate applicants based on predefined admission criteria.
3. To check academic percentages and entrance examination scores.
4. To identify eligible and ineligible applicants.
5. To reduce manual errors in admission processing.
6. To save time and improve processing efficiency.
7. To generate clear and consistent admission decisions.
8. To identify incomplete or invalid application details.
9. To maintain applicant information and decision records.
10. To support transparent and rule-based admission processing.

5.Assumptions

1. Each applicant has a unique application ID.
2. Applicant details are entered correctly.
3. Academic percentages are calculated using the applicable university rules.
4. Entrance examination scores are available for evaluation.
5. Admission criteria are predefined by the university.
6. The minimum eligibility requirements remain fixed during a processing session.
7. Missing or invalid information is flagged for verification.

8. The system evaluates applicants according to the configured rules.
9. Final admission is subject to document verification and applicable university regulations.
10. The system does not replace official admission authorities.

6.Algorithm

Step 1: Start.

Step 2: Enter the applicant ID and personal details.

Step 3: Enter the academic percentage and entrance examination score.

Step 4: Check whether the entered information is valid.

Step 5: Verify whether the applicant has completed the qualifying examination.

Step 6: Check whether the academic percentage is at least 50%.

Step 7: Check whether the entrance examination score is at least 60.

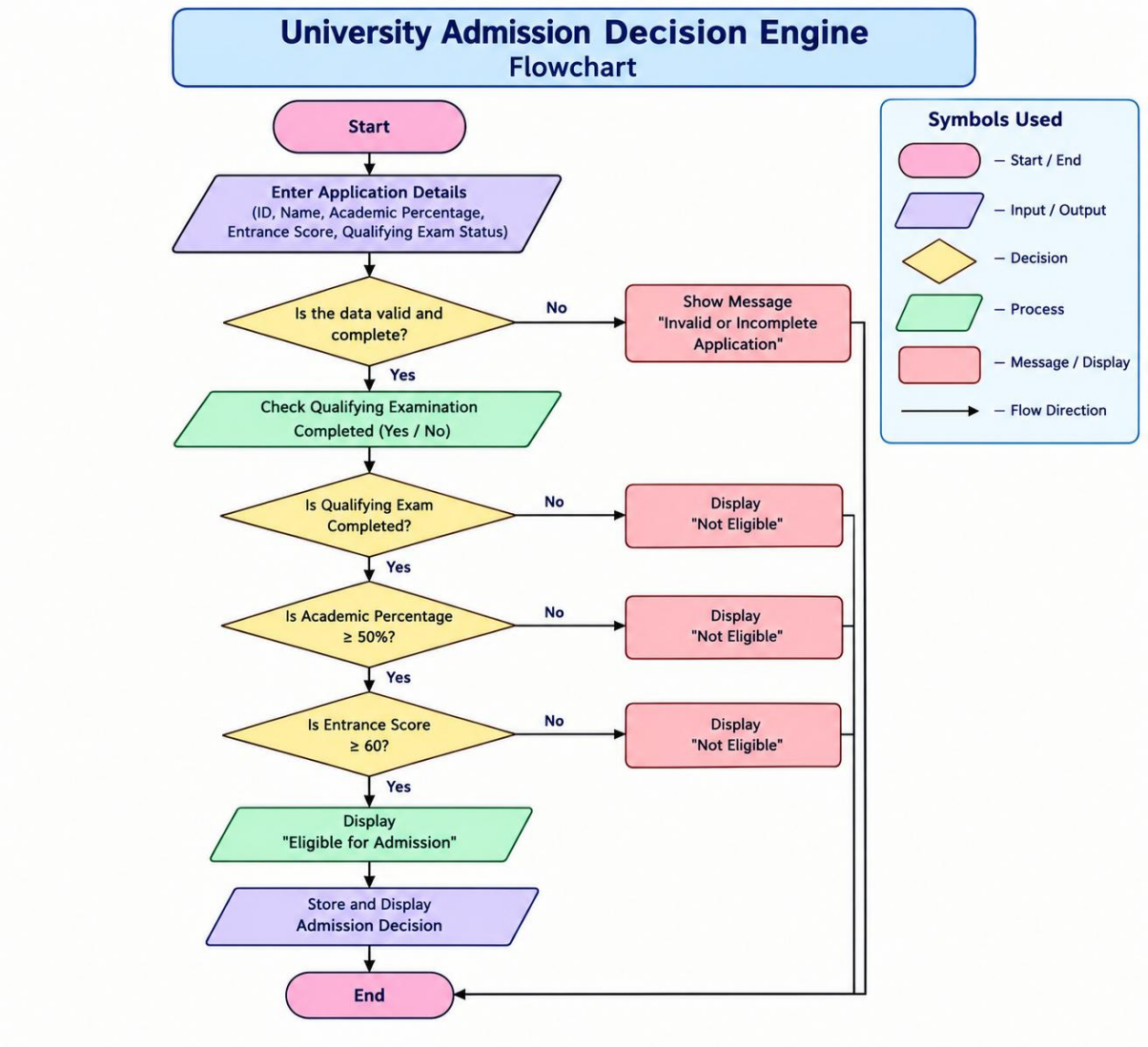
Step 8: If all conditions are satisfied, display "Eligible for Admission."

Step 9: Otherwise, display "Not Eligible."

Step 10: Store and display the admission decision.

Step 11: Stop.

7.Flowchart



8.Pseudocode

START

INPUT Applicant_ID

INPUT Academic_Percentage

INPUT Entrance_Score

INPUT Qualifying_Exam_Status

IF Academic_Percentage < 0 OR Academic_Percentage > 100

PRINT "Invalid Academic Percentage"

```
ELSE IF Entrance_Score < 0 OR Entrance_Score > 100
   PRINT "Invalid Entrance Score"

ELSE IF Qualifying_Exam_Status = "No"
   PRINT "Not Eligible"

ELSE IF Academic_Percentage >= 50 AND Entrance_Score >= 60
   PRINT "Eligible for Admission"

ELSE
   PRINT "Not Eligible"

ENDIF

STOP
```

9.Source Code

```
# University Admission Decision Engine

print("========================================")
print("   UNIVERSITY ADMISSION DECISION ENGINE")
print("========================================")

applicant_id = input("Enter Applicant ID: ")

academic_percentage = float(
   input("Enter Academic Percentage (0-100): ")
)
```

```
entrance_score = float(
    input("Enter Entrance Exam Score (0-100): ")
)

qualifying_exam = input(
    "Qualifying Exam Completed? (Yes/No): "
).strip().lower()

print("\n---------- ADMISSION RESULT ----------")

# Validate input values
if not 0 <= academic_percentage <= 100:
    print("Invalid Academic Percentage!")

elif not 0 <= entrance_score <= 100:
    print("Invalid Entrance Exam Score!")

elif qualifying_exam not in ("yes", "no"):
    print("Invalid Qualifying Exam Status!")

# Check admission eligibility
elif qualifying_exam == "no":
    print("Applicant ID:", applicant_id)
    print("Decision: NOT ELIGIBLE")
    print("Reason: Qualifying examination not completed.")

elif academic_percentage >= 50 and entrance_score >= 60:
    print("Applicant ID:", applicant_id)
```

```
    print("Academic Percentage:", academic_percentage)
    print("Entrance Exam Score:", entrance_score)
    print("Decision: ELIGIBLE FOR ADMISSION")

else:
    print("Applicant ID:", applicant_id)
    print("Academic Percentage:", academic_percentage)
    print("Entrance Exam Score:", entrance_score)
    print("Decision: NOT ELIGIBLE")

    if academic_percentage < 50:
        print("Reason: Academic percentage is below 50%.")

    if entrance_score < 60:
        print("Reason: Entrance score is below 60.")

print("----------------------------------------")
print("Admission Evaluation Completed.")
```

10.Test Cases

Test Case ID	Academic Percentage	Entrance Score	Qualifying Exam Completed	Expected Output
TC01	75%	80	Yes	Eligible for Admission
TC02	50%	60	Yes	Eligible for Admission
TC03	45%	80	Yes	Not Eligible
TC04	75%	55	Yes	Not Eligible
TC05	40%	50	No	Not Eligible
TC06	100%	100	Yes	Eligible for Admission

Test Case ID	Academic Percentage	Entrance Score	Qualifying Exam Completed	Expected Output
TC07	-5%	80	Yes	Invalid Academic Percentage
TC08	105%	80	Yes	Invalid Academic Percentage
TC09	75%	110	Yes	Invalid Entrance Exam Score
TC10	75%	80	No	Not Eligible
TC11	75%	80	Maybe	Invalid Qualifying Exam Status
TC12	0%	0	Yes	Not Eligible

11.Output Screenshots

```
----------------------------------------
   UNIVERSITY ADMISSION DECISION ENGINE
========================================
Enter Applicant ID: A101
Enter Academic Percentage (0-100): 75
Enter Entrance Exam Score (0-100): 80
Qualifying Exam Completed? (Yes/No): Yes

---------- ADMISSION RESULT ----------
Applicant ID: A101
Academic Percentage: 75.0
Entrance Exam Score: 80.0
Decision: ELIGIBLE FOR ADMISSION
----------------------------------------
Admission Evaluation Completed.

================ RESTART: C:\Users\VARSHINI\Desktop\python\activity2.py ========
========================================
   UNIVERSITY ADMISSION DECISION ENGINE
========================================
Enter Applicant ID: A102
Enter Academic Percentage (0-100): 45
Enter Entrance Exam Score (0-100): 55
Qualifying Exam Completed? (Yes/No): Yes

---------- ADMISSION RESULT ----------
Applicant ID: A102
Academic Percentage: 45.0
Entrance Exam Score: 55.0
Decision: NOT ELIGIBLE
Reason: Academic percentage is below 50%.
Reason: Entrance score is below 60.
----------------------------------------
Admission Evaluation Completed.
```

12.Conclusion

The University Admission Decision Engine provides a simple and efficient way to evaluate student applications based on predefined admission criteria. It checks academic percentages, entrance examination scores, and qualifying examination status to determine admission eligibility. The system reduces manual work, minimizes errors, and improves the speed and accuracy of the admission process. It also validates applicant information and displays appropriate messages for invalid inputs. Thus, this project provides a reliable solution for automating the university admission eligibility evaluation process.